

Quantum field theory as a theorem machine

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- ▶ I've gotten old enough that I regularly make cultural references to before any of you were born.
- ▶ I'm a mathematician, not a physicist.



I ask that myself sometimes.

And yet, every time I thought I was just doing mathematics, physics kept rearing its ugly head.

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Eventually it reached a point where I couldn't ignore it. As they say, if you can't beat 'em, join 'em.

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I thought that what I was working on ≈ 20 years ago was **almost commutative algebras**: non-commutative algebras that became commutative in a limit.

In physics terms, you would think about these in terms of a star-product deforming the functions on the phase space of a classical system.

But something special is happening here: since I'm working in an algebraic setting, I can think of these as the holomorphic functions on a complex manifold.

Taking the classical limit gives me a **hyperkähler** structure on this manifold.

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- ▶ All of the examples I was studying occurred this way!
- ▶ If you have a 3d theory has $\mathcal{N} = 4$ supersymmetry, it has two natural classes of topological twists $\mathcal{Q}_A$ and $\mathcal{Q}_B$, and thus two hyperkähler manifolds associated to it: the **Coulomb** and **Higgs** branches.

There is a particularly natural 3d gauge theory $\mathcal{T}(G, N)$ for a complex representation N of a complex reductive group G . In this case, the two branches are not created equal:

- ▶ $\mathfrak{M}_H$ is classical: the hyperkähler quotient $\mathbb{H}^n // G_c$ (Hitchin–Karlhede–Lindström–Roček, 1987). Mathematicians digested this decades ago.
- ▶ $\mathfrak{M}_C$ receives quantum corrections from monopoles. For ~ 20 years, physicists computed examples of a space with **no mathematical definition**.

Braverman–Finkelberg–Nakajima (2016): a rigorous definition, but quite technical and difficult to understand. I should probably emphasize: this definition was new to physicists!

A number of my research projects in recent years have focused on understanding this definition and its consequences more deeply.

3d mirror symmetry (Intriligator–Seiberg, 1996)

pairs of theories $\mathcal{T}, \mathcal{T}^\dagger$ equivalent in a way that swaps $\mathcal{Q}_A \leftrightarrow \mathcal{Q}_B$. In particular:

$$\mathfrak{M}_H(\mathcal{T}) \cong \mathfrak{M}_C(\mathcal{T}^\dagger) \quad \mathfrak{M}_C(\mathcal{T}) \cong \mathfrak{M}_H(\mathcal{T}^\dagger)$$

To a physicist: a duality. To a mathematician: a seemingly inexplicable pairing spaces with no visible relationship.

Despite the difficulty of understanding this connection, with collaborators, we had guessed a number of examples of dual pairs, and proved some of the consequences of this duality.

Chern–Simons theory:

- ▶ Defined by a “sum over histories” (path integral) where the notion of history is a connection on a principal G -bundle over a 3-manifold M .
- ▶ Given a knot $K \subset M$ and a representation V of G , we can define the Wilson loop observable $W_{K,V}$, which is a expected value of the trace of the holonomy of the connection around K in the representation V .

Witten (1989): expectation values of Wilson loops in Chern–Simons theory are polynomials in a parameter q ; $SU(2)$ with the fundamental representation gives the Jones polynomial.

Question

How does the invariant for $\bigwedge^m \mathbb{C}^N$ over $SU(N)$ vary as a function of m, N, q ?

There is expected to be a recursion in m and N (the **quantum $\hat{A}$ -polynomial**).

Physics:

- ▶ should be particularly simple in the **planar ('t Hooft) limit**:

$$N \rightarrow \infty, \quad q \rightarrow 1, \quad g = q^N \text{ fixed}$$

- ▶ should have a holographic dual description as a topological string on the resolved conifold, with a brane on the conormal Lagrangian of the knot.

This dual story has also been studied mathematically, and has its own invariants: the **augmentation variety** of the knot, defined by counting holomorphic curves with boundary on the brane. This is a Lagrangian subvariety of $(\mathbb{C}^*)^2$ (or $(\mathbb{C}^*)^3$ if we include g).

Thus we have two independently defined invariants of a knot: the augmentation variety, and the planar limit of the quantum $\hat{A}$ -polynomial. Aganagić–Vafa (2012) predicted that these are equal.

Idea (after Gaiotto–Kannagi–Sanjurjo): treat the coloring m as a **formal variable**. While this does not make sense as a representation of $SU(N)$, we can still make skein theoretic calculations.

For the unknot, substituting $g = q^N$, $\mu = q^m$ (no N , no $m!$):

$$\text{Unknot}(m) = \frac{q - q^{-1}}{\mu - \mu^{-1}} m - 1 \text{ (crossing)} m = \frac{qg\mu^{-1} - q^{-1}g^{-1}\mu}{\mu - \mu^{-1}} \lambda \text{ (crossing)} \text{Unknot}(m).$$

So $\mu - \mu^{-1} - \lambda(qg\mu^{-1} - q^{-1}g^{-1}\mu)$ annihilates the unknot: its deformed $\hat{A}$ -polynomial, with no computation of any knot invariant.

Theorem (W.–Zaimi, 2026)

At $q = 1$, the HOMFLYPT difference module **is** the degree-0 abelianized knot contact homology of K .

- ▶ The proof is purely algebraic: diagram calculus. No holomorphic curves, no path integrals. Elementary tools, non-elementary conclusion.
- ▶ First explicit mathematical connection between these two worlds.

This paper owes a large debt to conversations with Davide Gaiotto—the kind of accident that only happens in this building.

Just two examples of these sorts of connections:

- ▶ Physics predicted equalities between independently defined objects: Higgs $\leftrightarrow$ Coulomb; recursions $\leftrightarrow$ augmentation varieties.
- ▶ Proving them required inventing structures that now have lives of their own.

Physics and mathematics have a few centuries of profitable interaction behind them, and yet we keep finding new ways to surprise each other.